Session 4.4: Ethical Considerations in Artificial Intelligence: Algorithmic Bias, Regulatory Compliance, and AI Governance

TODAY'S PURPOSE

Why This Session Matters

- Check whether an AI model treats groups fairly
- Explain how the model makes decisions
- Report results to leaders who govern AI

Today's flow: 60 min lecture, then 120 min hands-on lab

EXPECTED OUTCOMES

By the end of today, you will be able to:

1. Measure disparate impact and demographic parity
2. Explain decisions with SHapley Additive exPlanations (SHAP)
3. Calibrate thresholds to meet the four-fifths rule

Concepts for This Session

CONCEPT 1

Algorithmic Disparate Impact

A legal and quantitative metric occurring when an automated decision system yields substantially different adverse outcomes across protected demographic groups.

CONCEPT 2

Demographic Parity Metric

The fairness standard requiring that the likelihood of receiving a positive algorithmic outcome (e.g., credit approval) is equal across all demographic groups.

CONCEPT 3

Model Explainability (SHAP / Feature Attribution)

Algorithmic frameworks that decompose complex black-box model predictions into transparent, auditable feature contributions for adverse action notices.

CONCEPT 4

Regulatory Compliance (EU AI Act Framework)

Auditing machine learning systems against emerging international governance standards (such as the European Union Artificial Intelligence Act for high-risk credit and hiring systems).

How the Concept Works in Practice

FOUNDATION 1

Sources of Algorithmic Bias in Historical Training Data

Machine learning models do not learn ethics; they learn historical correlation patterns. If historical lending or hiring practices contained human bias, models trained on that data will codify and scale that discrimination with algorithmic efficiency.

FOUNDATION 2

Quantitative Fairness Metrics (The Impossibility Theorem)

Understanding mathematical fairness trade-offs. It is mathematically impossible to satisfy Demographic Parity (equal approval rates), Equalized Odds (equal false positive rates), and Predictive Parity (equal accuracy) simultaneously when base demographic rates differ.

FOUNDATION 3

The European Union Artificial Intelligence Act (EU AI Act)

High-risk automated systems (credit underwriting, hiring, insurance pricing, critical infrastructure) face strict legal requirements: mandatory technical documentation, human-in-the-loop oversight, data quality governance, and adverse action explanations.

FOUNDATION 4

Establishing Boardroom AI Governance Frameworks

Corporate directors and C-suite executives bear fiduciary responsibility for algorithmic liabilities. Organizations must maintain automated audit trails, model risk registries, and independent red-team audits to protect enterprise reputation.

Build, Test, and Diagnose in the IDE

[1] FILE EXPLORER

Organize Assets

Use the session dataset and keep scripts in a clear workspace.

[2] CODE EDITOR

Write Small Segments

Run each completed segment before expanding the program.

[3] AI AGENT CHAT

Diagnose Errors

Capture the traceback, request an explanation, and review the fix.

[4] INTEGRATED TERMINAL

Verify Execution

Confirm the final program completes with return code 0.

Script Demonstration 1: Ethical Considerations in Artificial Intelligence: Algorithmic Bias, Regulatory Compliance, and AI Governance

```
import pandas as pd

    df = pd.read_csv("data/session_04_04_dataset.csv")

    # Approval rate by demographic group (Threshold >= 0.70)
    df["Approved_Binary"] = (df["Algorithmic_Approval_Score"] >=
0.70).astype(int)

    approval_rates = df.groupby("Demographic_Group")
["Approved_Binary"].mean()
    disparate_impact_ratio = approval_rates["Group_B"] /
approval_rates["Group_A"]

    print("Approval Rates by Demographic Group:\\n", approval_rates)
    print(f"Disparate Impact Ratio (Group B / Group A):
{disparate_impact_ratio:.3f}")
    # Four-Fifths (80%) Rule Compliance
    compliance = "Compliant (>= 0.80)" if disparate_impact_ratio >=
0.80 else "Non-Compliant Violation (< 0.80)"
    print(f"Regulatory Four-Fifths Rule Status: {compliance}")
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 2: Ethical Considerations in Artificial Intelligence: Algorithmic Bias, Regulatory Compliance, and AI Governance

```
def generate_adverse_action_notice(applicant_row):
        reasons = []
        if applicant_row["Credit_Score_Count"] < 700:
            reasons.append("Credit score below prime benchmark (700)")
        if applicant_row["Debt_To_Income_Pct"] > 30.0:
            reasons.append("Debt-to-income ratio exceeds underwriting
ceiling (30%)")
        return "; ".join(reasons) if reasons else "Approved"

    df["Regulatory_Explanation"] =
df.apply(generate_adverse_action_notice, axis=1)
    rejected_sample = df[df["Approved_Binary"] == 0].iloc[0]
    print(f"Adverse Action Notice for
{rejected_sample['Applicant_ID']}:
{rejected_sample['Regulatory_Explanation']}")
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 3: Ethical Considerations in Artificial Intelligence: Algorithmic Bias, Regulatory Compliance, and AI Governance

```
audit_record = {
        "Model_Name": "Credit_Underwriting_Risk_Engine_v1.4",
        "Total_Audited_Applicants": len(df),
        "Group_A_Approval_Rate": float(approval_rates["Group_A"]),
        "Group_B_Approval_Rate": float(approval_rates["Group_B"]),
        "Disparate_Impact_Ratio": float(disparate_impact_ratio),
        "Governance_Status": "Audited_Board_Review_Required"
    }
    print("Governance Audit Certificate:", audit_record)
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Exercise 1 and Exercise 2

EXERCISE 1: GUIDED

Foundational Practice

Follow the lab protocol, run the script, and verify its output with the session dataset.

EXERCISE 2: DEBUGGING

Applied Challenge

Extend the program, diagnose an intentional error, apply the verified correction, and confirm return code 0.